**6** A force of 70 newtons acts on an area of 20 $cm^2$

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The force is increased by 10 newtons.
The area is increased by 10 $cm^2$

Helen says,

"The pressure decreases by less than 20%"

Is Helen correct?
You must show how you get your answer.

A = area in $cm^2$; F = force in N.

| | A | F | |
|---|---|---|---|
| | 20 | 70 | |
| ÷20 | | | ÷20 |
| | 1 | 3.5 | |

Old pressure = 70 ÷ 20 = 3.5 $N/cm^2$

**1 mark**
Find a pressure using force ÷ area

New force = 80 N; new area = 30 $cm^2$.

$$\text{New pressure} = \frac{80}{30} = \frac{8}{3} \approx 2.6667 \text{ } N/cm^2$$

Pct = percentage; p = old pressure in $N/cm^2$.

| | Pct | p | |
|---|---|---|---|
| | 100 | 3.5 | |
| ÷10 | | | ÷10 |
| | 10 | 0.35 | |
| ×2 | | | ×2 |
| | 20 | 0.7 | |

20% of 3.5 = 0.7

After 20% decrease: 3.5 − 0.7 = 2.8

**1 mark**
Compare with a 20% decrease

2.6667 < 2.8, so the decrease is more than 20%.

No, Helen is not correct.

**1 mark**
Correct, supported decision

**(Total for Question 6 is 3 marks)**